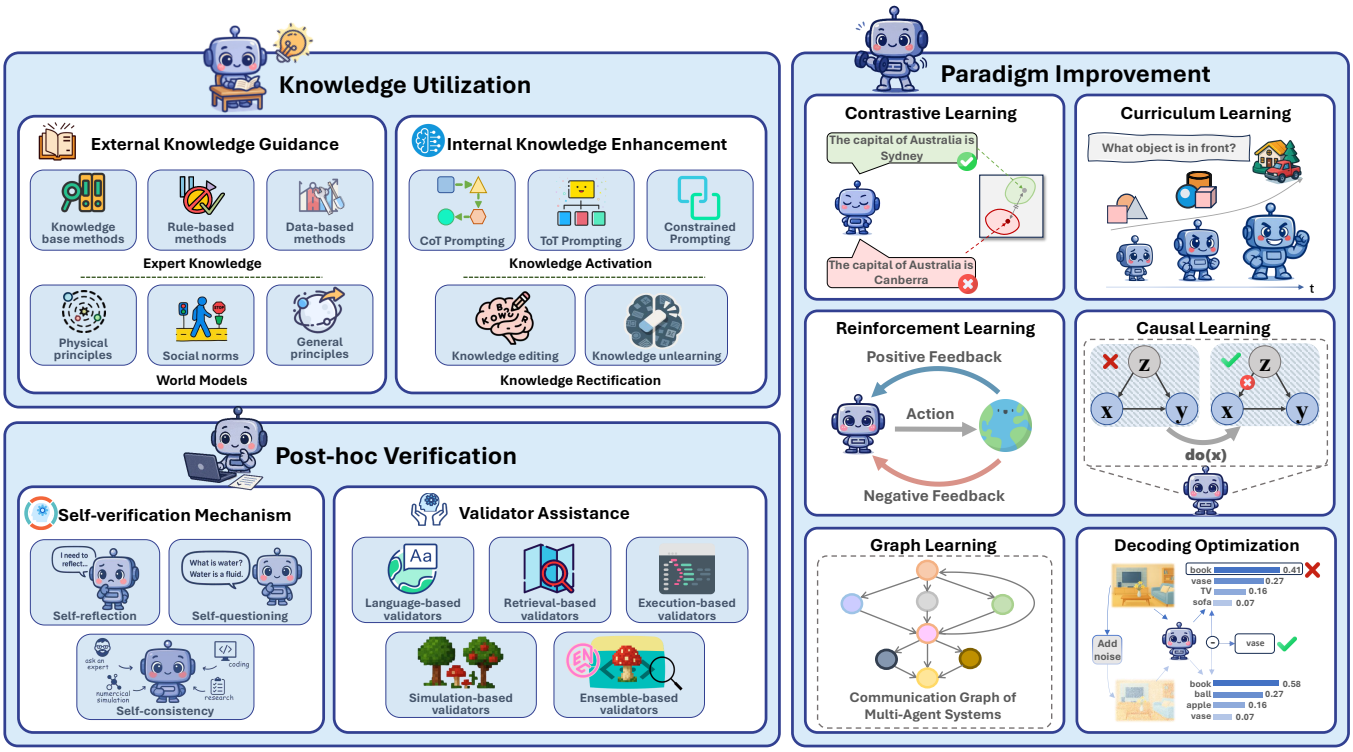

Fig. 3. A simple illustration of approaches to agent hallucination mitigation. It encompasses three branches, knowledge utilization, paradigm improvement, and post-hoc certification, comprising a total of ten representative methods.

meta-learning approach [184], which aims to to learn how to update the model's parameters for adjusting knowledge for new information. The second is **Knowledge Unlearning**. In contrast to knowledge editing, it focuses on removing erroneous knowledge to improve the agent's reliability. It includes two main approaches [185]: parameter optimization [186], which fine-tunes model parameters under certain conditions (e.g., parameter update scopes[187]) to forget some targeted knowledge; parameter merging [188], which performs offline operations on model parameters without further training.

2) Paradigm Improvement. This section focuses on enhancing the agent's capabilities through general learning paradigms to mitigate hallucinations. We summarize six representative paradigms as follows.

a) Contrastive Learning (CoL) is a well-established self-supervised learning paradigm whose core idea is to learn more discriminative representations by comparing similarities and differences between samples [189], [190], [191]. By incorporating contrastive learning, the agent can better identify relevant and accurate patterns, reducing errors when handling unfamiliar or incomplete task inputs and effectively mitigating hallucination issues [192], [193], [194].

b) Curriculum Learning (CuL) is a learning paradigm inspired by the way humans learn. Its core idea is to start training the model with "easy" examples and gradually move to "harder" ones, improving learning efficiency and generalization ability [189]. By leveraging curriculum learning, agents rapidly accumulate successful experiences by starting with simpler tasks and gradually transferring this knowledge to more complex scenarios. This progressive learning paradigm enhances their fundamental capabilities, effectively mitigating agent hallucinations [195], [196], [94].

c) Reinforcement Learning (RL) serves as a fundamental learning paradigm, emulating the biological learning process of "trial, feedback, and adjustment". This approach enables agents to interact with their environment and learn to maximize cumulative rewards through iterative trial and error [197]. Through the reinforcement learning paradigm, the agent can adjust its policy based on rewards, thereby optimizing future decision-making, learning to achieve goals in complex environments, and significantly mitigating potential hallucination issues in multi-turn interactions [198], [199], [200], [201], [202]. In addition, there are also studies that aim to improve the reliability of reward signals in order to enhance the overall model effectiveness [203].

d) Causal Learning (CaL) refers to the process in statistical machine learning that goes beyond capturing mere correlations and instead aims to model, discover, and leverage causal relationships between variables, thereby enhancing the model's interpretability and generalization ability [204], [205]. In the context of LLM-based agents, identifying and leveraging potential causal relationships within task inputs and decision-making processes enables more accurate understanding of how different factors influence task outcomes. This helps reduce representational errors and logical inconsistencies for mitigating perception [206], [207] and reasoning hallucinations [208], [209].

e) Graph Learning (GL) is a class of machine learning

TABLE I
AN OVERVIEW OF MITIGATION METHODS. TO SAVE SPACE, WE USE THE ABBREVIATIONS OF HALLUCINATION TYPES AND MITIGATION METHODS MENTIONED EARLIER. "✔" INDICATES THAT RELATED WORKS ALREADY EXIST; "–" INDICATES THAT IT HAS NOT YET BEEN EXPLORED.

Type \ Branch		Knowledge Utilization		Paradigm Improvement						Post-hoc Verification	
		EKG	IKE	CoL	CuL	RL	CaL	GL	DO	SM	VA
Reasoning Hallucinations	GUHs	✔	✔	✔	–	✔	–	–	–	✔	✔
	IDHs	✔	✔	–	–	–	–	✔	✔	✔	✔
	PGHs	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
Execution Hallucinations	TSHs	✔	✔	✔	✔	✔	–	✔	–	✔	✔
	TCHs	✔	✔	✔	✔	✔	–	–	–	✔	✔
Perception Hallucinations		✔	✔	✔	–	✔	✔	–	✔	–	✔
Memorization Hallucinations	MRHs	✔	–	–	–	–	–	✔	–	–	✔
	MUHs	✔	✔	✔	–	–	–	✔	–	–	✔
Communication Hallucinations		✔	–	–	–	✔	–	✔	–	✔	✔

paradigms specifically designed to handle graph-structured data, which has been widely applied in various domains such as social networks, bioinformatics, and recommendation systems [210], [211], [212]. It enhances the agent's capability to organize and manage tasks in a structured and systematic manner, particularly for tool usage and memory updates, thereby reducing the risk of agent hallucinations [213], [17], [214]. For example, ControlLLM [215] builds a tool graph comprising resource and tool nodes, and employs a parallel graph search algorithm to efficiently find appropriate tool paths for mitigating execution hallucinations. More importantly, recent studies leverage graph learning to optimize communication topology in LLM-based MAS [157], [216], [217]. By dynamically learning which agents should communicate, when, and through what channels, the system can suppress noisy message passing while promoting informative interactions.

Unlike the aforementioned training-phase learning paradigms, *f) Decoding Optimization (DO)* is a test-time learning paradigm. This paradigm can improve output quality by adjusting probability distributions or attention patterns in decoding processes, ensuring better alignment with inputs and factual knowledge,thereby reducing hallucinations stemming from reasoning errors [54], [218], [219]. Common decoding strategies include contrastive decoding, self-calibrated attention decoding [220] and fusion decoding [218]. Contrastive decoding [221] selects more reliable and context-consistent outputs by comparing multiple decoding candidates. Self-calibrated attention decoding [220] aims to dynamically adjust the attention weights to capture more informative cues. Fusion decoding incorporates the multi-source information to reduce dependence on single-modal priors for mitigating hallucinations. CGD [222] is one of representative works, which integrates semantic similarity scores between textual and visual content with final-layer token likelihoods to reduce reliance on language priors.

3) Post-hoc Verification. This paradigm focuses on monitoring and evaluating outputs after task execution. It generally follows a multi-step process in which, immediately after each step, the agent assesses the validity, consistency, and factuality of the intermediate decisions and actions generated in that specific step. This assessment helps prevent the accumulation and propagation of hallucinations in long-horizon tasks. Existing approaches can be broadly categorized into two types: **a) Self-verification Mechanism**, in which the agent introspectively reviews its own behaviors using an internal reasoning strategy or prompt-based self-assessment. **b) Validator Assistance**, which leverages another independent system to detect potential flaws in the agent's outputs.

a) Self-verification Mechanism (SM). Self-verification is a lightweight and model-internal approach wherein agents assess the validity and reliability of their own outputs without relying on external validators. It plays a crucial role in hallucination mitigation [223], and several technical methods have been explored to implement self-verification. Specifically, **Self-reflection** enables agents to revisit and critique their own outputs, often through prompting techniques that encourage introspection and identification of reasoning flaws [89], [224], [15], which can be further facilitated by estimating the agent's own confidence or uncertainty [225], [226], [50]. **Self-consistency** leverages the generation of multiple candidate outputs, such as diverse reasoning paths or answers, and aggregates them using majority voting or confidence-weighted schemes to select the most reliable results [177], [81], [78]. **Self-questioning** guides the agent to pose and answer critical verification questions grounded in its own reasoning process, enabling the detection of unsupported assertions [227], [228], [74]. Collectively, these strategies establish a foundation for autonomous error detection and correction, empowering agents to self-regulate hallucination risks.

b) Validator Assistance (VA). This approach leverages external validators to verify the correctness of an agent's outputs, aiming to mitigate hallucinations [141], [142]. According to the types of external validators, we divide existing methods into the following five categories: **Language-based Validators** independently assess the truthfulness or coherence of an agent's outputs using techniques such as atomic fact

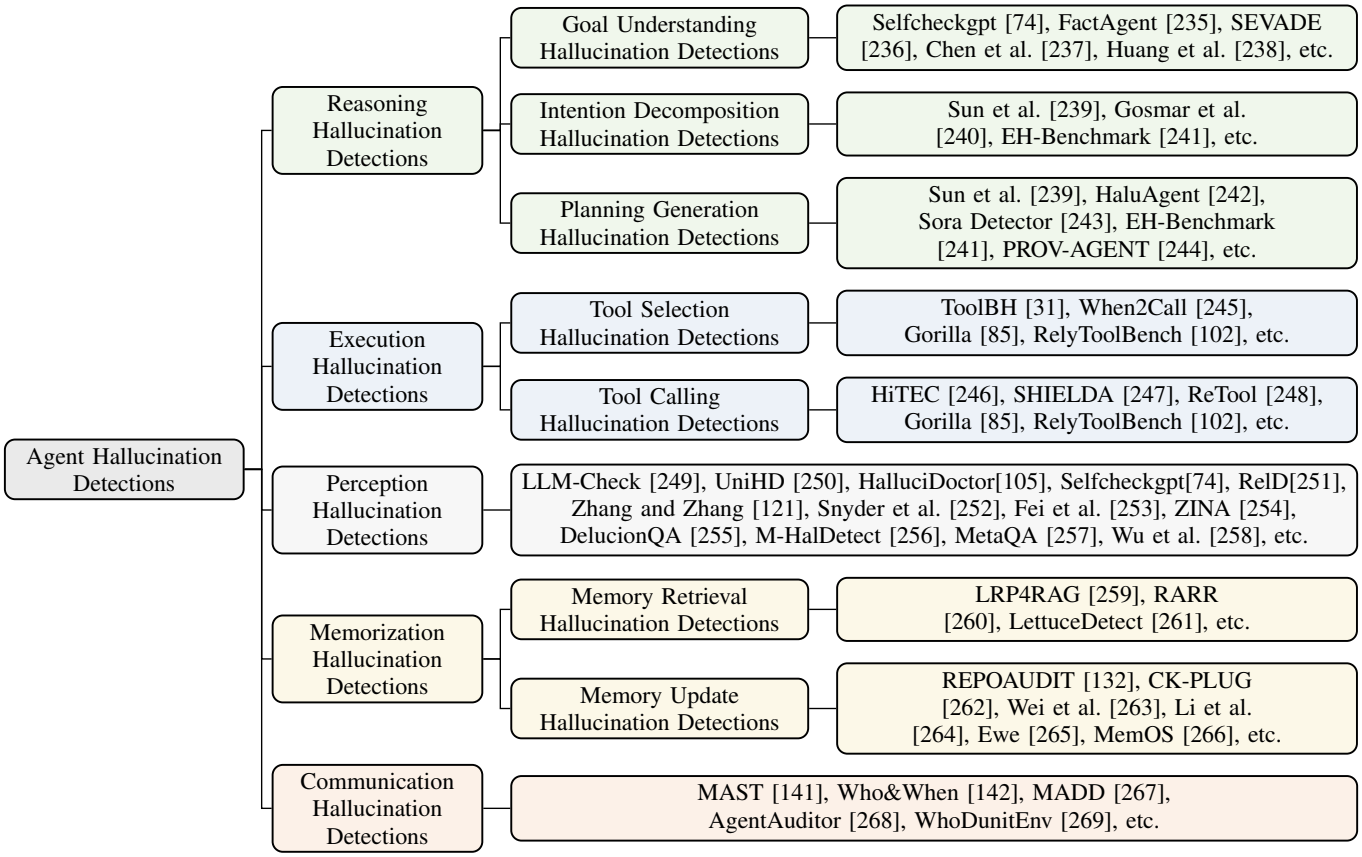

Fig. 4. A typology of methods of agent hallucination detection. We highlight the representative approaches for each type of agent hallucinations.

decomposition and entailment checking [229], [230], [231]. **Retrieval-based Validators** rely on some external sources such as search engines to verify whether outputs aligns with established facts [232]. **Execution-based Validators** evaluate outputs by running generated codes or plans in external execution environments, enabling direct assessment of correctness through functional outcomes [233]. **Simulation-based Validators** validate agent behaviors through interaction with sandboxed environments, allowing for realistic testing in tasks involving embodiment, planning, or sequential control [234]. Building on these categories, an increasing number of studies have explored **Ensemble-based Validators** that integrate multiple types of validators to improve robustness. By enabling cross-verification among different approaches, these methods help mitigate the limitations inherent in individual validation strategies [78]. In the end, Table I summarizes which types of agent hallucinations can be addressed by the ten listed mitigation methods.

B. Agent Hallucination Detection

Based on the introduced taxonomy of agent hallucinations, we summarize the existing detection methods in Fig. 4. From it, we can observe that unlike the mitigation strategies discussed above, research on agent hallucination detection remains relatively limited. Among these detection methods, those addressing perception hallucinations are relatively numerous, whereas methods on memorization hallucinations and communication hallucinations are comparatively limited. We believe this is because perception belongs to the shallow layers of the agent, making hallucination detection and error identification relatively straightforward, which explains the larger amount of corresponding work. In contrast, memory and communication are part of the deeper layers of the agent, where the final outputs are coupled with computations from numerous intermediate modules. This makes hallucination detection and localization more challenging, and consequently, there is comparatively less work in these areas.

V. Future Directions

LLM-based agent hallucinations represent an emerging research frontier that has attracted increasing attention from both academia and industry. As discussed above, a substantial body of work has focused on the mitigation and detection of agent hallucinations. Building upon our established taxonomy and existing literature, this survey introduces several promising directions for future study.

1) Hallucinatory Accumulation Investigation. Most existing studies investigate hallucination instances and their underlying causes within a single agent loop. However, as we emphasize, agent decision-making is inherently a multi-step and sequential process, in which hallucinations can accumulate and amplify over time. In such cases, hallucinations may initially appear as minor issues, but their iterative accumulation can ultimately lead to severe consequences. Compared to single-step scenarios, hallucination accumulation presents a significantly more